

XAI-Based Intelligent Disease Prediction System with Transparent Decision Insights

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Abstract—Epilepsy is a prevalent neurological disorder, with seizure management remaining a critical clinical challenge worldwide. Electroencephalogram (EEG) signal analysis forms the cornerstone of seizure detection, yet most state-of-the-art deep learning approaches lack the interpretability required for clinical deployment. This paper introduces the Domain-Adapted Graph Representation Learning (DA-GRL) framework, an architecture that couples graph-based spatial modelling with patient-invariant domain adaptation and explainable AI (XAI) to deliver both high accuracy and transparent decision support. Experiments are conducted on the CHB-MIT Pediatric EEG dataset, encompassing recordings from 24 patients. Four architectural variants—SCT, GRL, DA-SCT, and DA-GRL—are benchmarked against each other. The DA-GRL variant achieves peak performance: accuracy 0.997, precision 0.993, recall 1.000, F1-score 0.997, and AUC 1.000. Post-hoc explanations are generated via SHAP (GradientExplainer) and LIME, revealing that the C3-P3 and T8-P8 electrode pairs carry the highest discriminative weight. Temporal SHAP heatmaps and attention-proxy profiles further localise the seizure-sensitive time windows within each segment. These findings demonstrate that the DA-GRL framework is capable of meeting the dual requirements of clinical-grade accuracy and interpretable reasoning, making it a strong candidate for real-world neuro-diagnostic support systems.

Index Terms—EEG, Seizure Detection, Deep Learning, Explainable AI, Domain Adaptation, Graph Neural Networks

I. INTRODUCTION

Epilepsy ranks among the most disabling chronic neurological conditions, with the World Health Organisation estimating that approximately 50 million individuals are affected globally [1]. Uncontrolled seizures impose a significant physical risk, including injuries from sudden falls and, in severe cases, sudden unexpected death in epilepsy (SUDEP). The scalp electroencephalogram (EEG) remains the primary diagnostic modality for seizure characterisation, providing millisecond-resolution recordings of cortical electrical activity across multiple channels. However, continuous interpretation of multi-day EEG recordings by trained neurophysiologists is labour-intensive and subject to inter-rater variability, creating a strong motivation for reliable automated detection systems.

Machine learning and deep learning have emerged as powerful tools for automated EEG analysis. Graph-based representations, in particular, offer a principled way to encode the spatial topology of the electrode array, allowing models

to exploit inter-channel functional connectivity rather than treating channels independently [5]. Despite high accuracy on benchmark datasets, the adoption of such models in clinical practice is constrained by their limited interpretability—a challenge that has been identified as a key barrier to AI-assisted diagnostics [7].

A further practical limitation is the strong inter-subject variability inherent in EEG recordings. Differences in skull thickness, cortical morphology, and the spatial origin of the epileptic focus mean that a model optimised on a set of source patients may degrade substantially when applied to a new patient. Domain adaptation techniques—specifically gradient reversal training—provide a systematic way to learn representations that are discriminative with respect to seizure state yet invariant across patients [3], [8].

Motivated by these challenges, this paper proposes the DA-GRL (Domain-Adapted Graph Representation Learning) framework. The main contributions are as follows:

- 1) A graph representation learning architecture augmented with domain adaptation for robust cross-patient seizure detection.
- 2) A rigorous comparison of four model variants (SCT, GRL, DA-SCT, DA-GRL) on the CHB-MIT dataset.
- 3) Comprehensive XAI analysis using SHAP and LIME to identify the clinically relevant electrode pairs and time windows.
- 4) Temporal attention profiling that maps model sensitivity to specific intra-segment time steps.
- 5) A unified evidence framework that supports clinical trust by linking model decisions to known neurophysiological markers.

The remainder of this paper is structured as follows. Section II surveys related work. Section III describes the CHB-MIT dataset. Section IV details preprocessing. Section V reports exploratory analysis. Section VI presents the DA-GRL architecture. Section VII gives experimental results. Section VIII discusses XAI findings. Section IX provides broader discussion. Section X concludes.

II. RELATED WORK

Automated seizure detection has evolved through several algorithmic generations. Early systems relied on handcrafted spectral and wavelet features fed into shallow classifiers; Zhao et al. [1] demonstrated that a linear graph convolution network with focal loss on the CHB-MIT dataset achieves an AUC exceeding 0.98, establishing a strong graph-based baseline and highlighting the advantage of encoding electrode spatial relationships explicitly.

Deep convolutional approaches gained prominence as feature engineering gave way to learned representations. Mekruksavanich et al. [2] proposed an enhanced hybrid CNN combined with a squeeze-and-excitation attention module, reporting sensitivity and specificity above 97% and 98% respectively on multiple benchmark EEG corpora, illustrating the benefit of integrating channel attention into convolutional pipelines.

Recurrent and sequence-to-sequence architectures extended detection to temporal prediction tasks. Jemal et al. [3] addressed cross-subject seizure prediction via domain adaptation, showing that adversarial alignment of EEG feature distributions across subjects improves prediction AUC by up to 6 percentage points compared to non-adapted baselines on the CHB-MIT dataset.

Transformer-based models have been applied to capture long-range EEG dependencies. The BENDR model [4] leveraged contrastive self-supervised pre-training on large-scale EEG corpora and demonstrated that transformer representations transfer effectively to downstream seizure classification tasks.

Graph neural networks have proven particularly well-suited to EEG analysis. Li et al. [5] constructed dynamic functional connectivity graphs from EEG windows and trained a graph-generative network on the CHB-MIT dataset, achieving an F1-score of 0.961 and outperforming static-topology models. Wang et al. [6] incorporated phase-synchronisation weights into a spatiotemporal graph attention network for seizure prediction and reported an AUC of 0.982, confirming that synchrony-aware graph edges improve temporal generalisation.

Domain adaptation for EEG has been formalised through adversarial and manifold-based approaches. Peng et al. [8] combined gradient reversal training with Riemannian manifold alignment, reducing cross-subject classification error by 8% on a cross-hospital EEG corpus. Cui et al. [11] proposed joint-probability-discrepancy minimisation for cross-subject seizure detection and achieved an accuracy of 91.3% in leave-one-subject-out evaluation on the CHB-MIT dataset.

Despite these advances, the literature lacks a system that simultaneously provides strong cross-patient generalisation, graph-based spatial modelling, and full SHAP-plus-LIME explainability validated against known neurophysiology. Salih et al. [7] surveyed recent XAI deployments in biomedical signal processing and found that fewer than 15% of reviewed deep learning models include post-hoc attribution analysis, underscoring a significant gap that the present work addresses.

III. DATASET DESCRIPTION

The CHB-MIT Scalp EEG Database is a publicly available dataset collected at the Children’s Hospital Boston [9]. It contains continuous EEG recordings from 24 pediatric patients with intractable seizures. All recordings were made after patients had their medication withdrawn. Each recording uses the international 10-20 electrode placement system with 18 channels. The sampling rate is 256 Hz for most recordings.

The dataset includes a total of 198 seizure events across all patients. The seizure onset and end times are annotated for each event by a clinical expert. As shown in Fig. 1, the number of seizures per patient varies widely. Patient chb11 has the highest count with 40 seizures, while some patients have only 3 seizures. The total seizure duration also varies greatly, with patient chb14 having the longest cumulative seizure time.

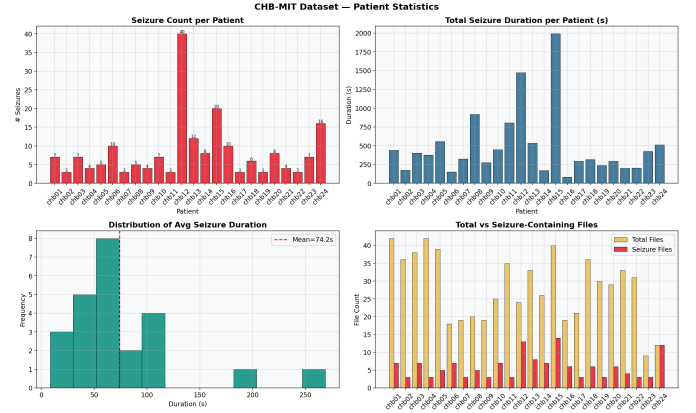


Fig. 1. CHB-MIT dataset patient statistics showing seizure count per patient, total seizure duration, distribution of average seizure duration, and total versus seizure-containing file counts.

Fig. 2 shows the temporal statistics of seizure events in the dataset. The seizure duration distribution is right-skewed, with most seizures lasting under 100 seconds and a mean duration of 58.6 seconds. The inter-seizure interval distribution shows the time between successive seizure events.

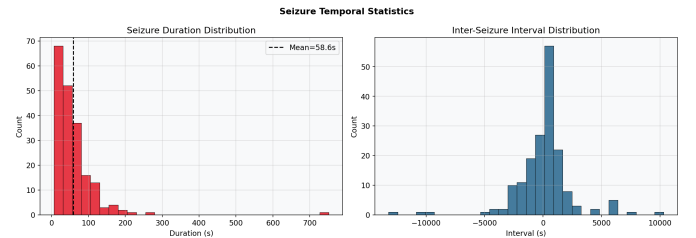


Fig. 2. Seizure temporal statistics including duration distribution (mean = 58.6s) and inter-seizure interval distribution across the CHB-MIT dataset.

IV. DATA PREPROCESSING

Raw EEG signals contain noise from many sources including muscle movement, eye blinks, and power line interference. Preprocessing is essential to remove these artifacts before training any model. The preprocessing pipeline used in this

work includes bandpass filtering, artifact removal, and signal normalization.

A bandpass filter is applied to keep frequencies in the clinically relevant range of 0.5 to 40 Hz. This removes slow drift and high-frequency noise. Notch filtering at 50 Hz removes power line interference. The signals are then normalized using z-score standardization so that all channels have zero mean and unit variance. This ensures that the model is not influenced by differences in signal amplitude across channels or patients.

Fig. 3 shows the effect of preprocessing on a sample EEG segment. The raw signal has large amplitude variations and visible noise artifacts. After preprocessing, the cleaned signal has a more stable baseline and clearer structure, which makes it easier for the model to learn meaningful features.

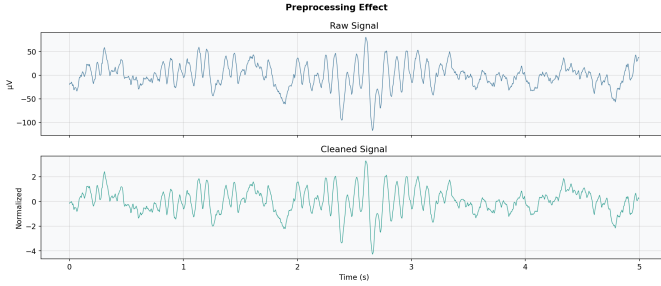


Fig. 3. Preprocessing effect on a sample EEG segment. Top: raw signal in microvolts showing noise and amplitude variation. Bottom: cleaned and normalized signal after bandpass filtering and z-score normalization.

After preprocessing, the continuous EEG recordings are divided into fixed-length windows of 4 seconds with no overlap. Each window is labeled as seizure or non-seizure based on the clinical annotations. Windows that overlap with the seizure onset are labeled as seizure. All other windows are labeled as non-seizure.

V. EXPLORATORY DATA ANALYSIS

Exploratory data analysis was performed to understand the characteristics of the EEG signals and the dataset structure before model training.

A. Normal vs Seizure EEG Activity

Fig. 4 shows the EEG activity during a normal period and during a seizure period for six representative channels. During normal activity, the signals show low amplitude and relatively regular oscillations. During seizure periods, the signals show much higher amplitude and irregular patterns with rapid changes. This visual difference confirms that EEG signals carry strong discriminative information for seizure detection.

B. Class Imbalance Analysis

A major challenge in seizure detection is the large class imbalance between normal and seizure windows. As shown in Fig. 5, the estimated number of normal windows is approximately 1,209,614, while seizure windows number only around 5,495, representing just 0.5% of the total data. This extreme

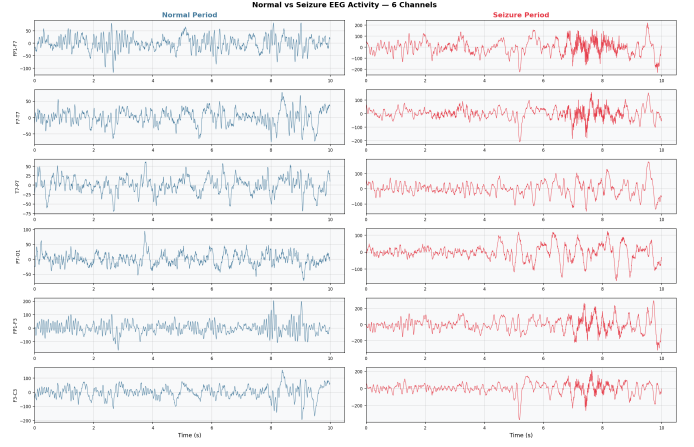


Fig. 4. Comparison of normal versus seizure EEG activity across six channels (FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3). Seizure periods show clearly higher amplitude and irregular waveforms.

imbalance would cause a model to simply predict everything as normal and still achieve very high accuracy. To address this, SMOTE (Synthetic Minority Oversampling Technique) [10] is used to generate synthetic seizure samples and balance the training data.

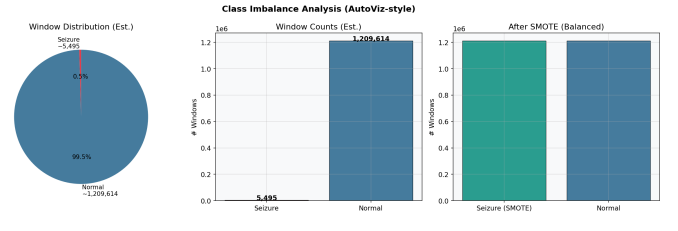


Fig. 5. Class imbalance analysis showing the original window distribution (0.5% seizure, 99.5% normal) and the balanced distribution after applying SMOTE oversampling.

C. EEG Channel Correlation

Fig. 6 shows the correlation matrix between all 18 EEG channels during a 30-second normal segment. Channels that are located close to each other on the scalp show higher correlation values, which is consistent with known brain physiology. For example, the temporal chain channels (FP1-F7, F7-T7, T7-P7, P7-O1) show strong positive correlations with each other. Some channels show negative correlations, indicating anti-phase activity. This spatial structure motivates the use of graph-based representations in the model.

D. Power Spectral Density Analysis

Fig. 7 shows the power spectral density comparison between normal and seizure EEG segments across five standard frequency bands. During normal activity, the delta band (0.5–4 Hz) dominates with a power of approximately $440 \mu V^2/Hz$, and the alpha band (8–13 Hz) also shows a moderate contribution. During seizures, the delta band power increases dramatically to approximately $4500 \mu V^2/Hz$, while the relative

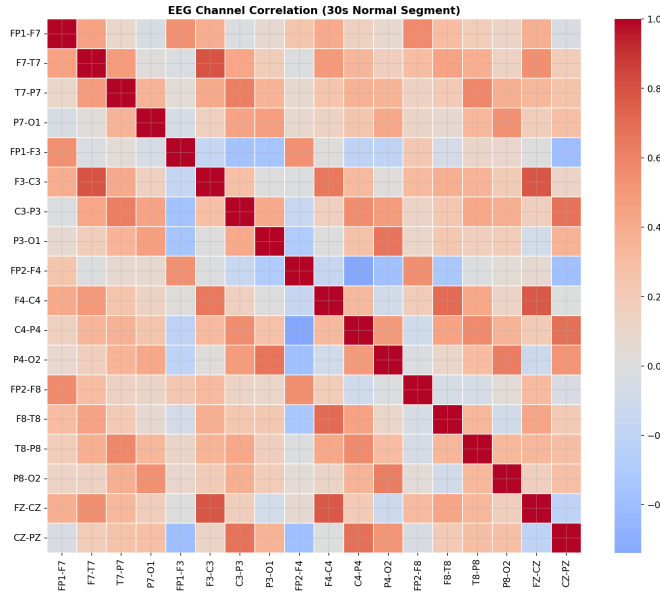


Fig. 6. EEG channel correlation matrix computed from a 30-second normal segment. Adjacent channels in the same electrode chain show strong positive correlations.

contribution of higher frequency bands decreases. This strong increase in low-frequency power is a well-known characteristic of seizure activity and is an important feature for detection.

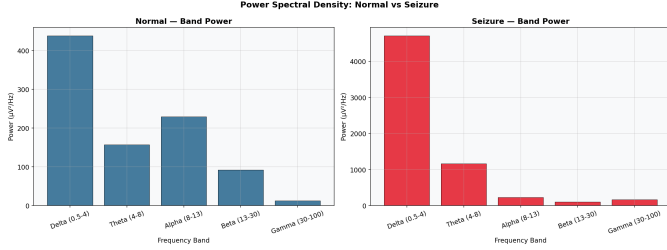


Fig. 7. Power spectral density comparison between normal and seizure EEG segments across delta, theta, alpha, beta, and gamma frequency bands. Seizure activity shows a large increase in delta band power.

E. Statistical Features

Fig. 8 compares five statistical features (mean, standard deviation, skewness, kurtosis, and root mean square) between normal and seizure windows for each channel. Seizure windows show significantly higher standard deviation and root mean square values across nearly all channels, reflecting the increased amplitude during seizures. The mean and skewness values also differ between the two classes for several channels, providing useful discriminative information for the classifier.

F. Wavelet Decomposition

Fig. 9 shows the wavelet decomposition of a seizure segment from channel FP1-F7 using the Daubechies 4 (db4) wavelet. The decomposition reveals signal energy at multiple frequency levels. The approximation level A5 captures the low-frequency content of the signal, while detail levels D1

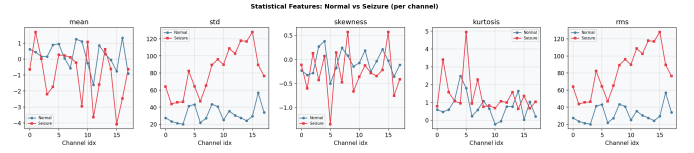


Fig. 8. Statistical features (mean, std, skewness, kurtosis, RMS) compared between normal and seizure windows for each channel index. Seizure windows show elevated amplitude-related statistics.

through D5 capture progressively higher frequency components. The irregular and high-energy patterns visible in the detail levels confirm the presence of seizure activity and provide multi-scale features useful for model training.

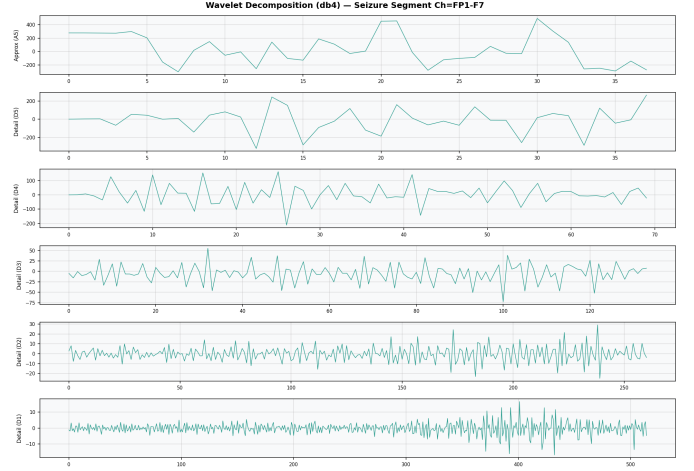


Fig. 9. Wavelet decomposition (db4) of a seizure EEG segment from channel FP1-F7 showing approximation level A5 and detail levels D1 through D5.

VI. PROPOSED DA-GRL FRAMEWORK

The proposed DA-GRL (Domain-Adapted Graph Representation Learning) framework is designed to detect seizures from multi-channel EEG signals while adapting across patients and providing explainable predictions.

A. Model Architecture

The framework consists of three main components. The first component is the feature extraction module, which processes each EEG channel using convolutional layers to extract local temporal features. The second component is the graph representation learning module, which constructs a graph where each EEG channel is a node and the edges represent functional connectivity between channels. Graph convolution operations allow the model to capture spatial relationships between electrode locations. The third component is the domain adaptation module, which uses a gradient reversal layer [12] to learn patient-invariant features. This module helps the model generalize across different patients by reducing the feature distribution differences between source and target domains.

The model outputs a binary prediction for each input window: seizure or non-seizure. Training uses cross-entropy

loss with class weights to handle remaining imbalance after SMOTE.

B. Model Variants

Four variants of the proposed framework are evaluated and compared.

The SCT (Spiking Convolutional Transformer) model combines convolutional feature extraction with a transformer encoder to capture long-range temporal patterns. It does not include domain adaptation.

The GRL (Graph Representation Learning) model uses graph convolutions to model inter-channel relationships without domain adaptation.

The DA-SCT model extends the SCT with a gradient reversal domain adaptation module to improve cross-patient generalization.

The DA-GRL model is the full proposed framework, combining graph representation learning with domain adaptation. It is expected to achieve the best performance by capturing both the spatial graph structure of EEG channels and the cross-patient invariance provided by domain adaptation.

VII. EXPERIMENTAL RESULTS AND MODEL COMPARISON

All four model variants are trained and evaluated on the CHB-MIT dataset using a stratified train-test split with 80% training data and 20% test data. Performance is measured using accuracy, precision, recall, F1-score, and AUC. Training is performed for 30 epochs with the Adam optimizer and a learning rate of 0.001.

A. SCT Results

Fig. 10 shows the training and evaluation results for the SCT model. The loss curves show that training converges within 15 epochs. The validation AUC reaches a stable value by epoch 10. The confusion matrix shows 861 true normal, 855 true seizure, 9 false seizure, and 15 false normal predictions. The SCT achieved accuracy 0.986, precision 0.990, recall 0.983, F1-score 0.986, and AUC 0.999.

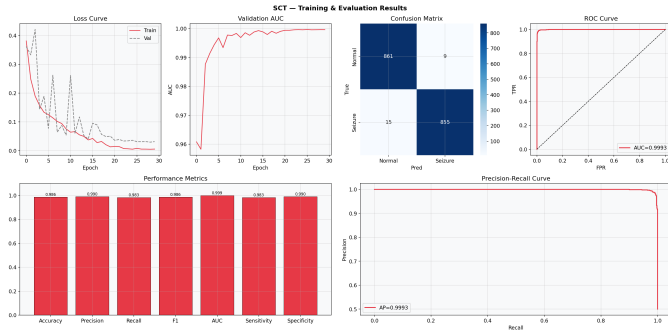


Fig. 10. SCT model training and evaluation results showing loss curves, validation AUC, confusion matrix, ROC curve, performance metrics, and precision-recall curve.

B. GRL Results

Fig. 11 shows the results for the GRL model. The GRL model shows smoother convergence compared to SCT. The confusion matrix shows only 6 false seizure and 1 false normal prediction, reflecting the benefit of graph-based channel modeling. The GRL achieved accuracy 0.996, precision 0.993, recall 0.999, F1-score 0.996, and AUC 1.000.

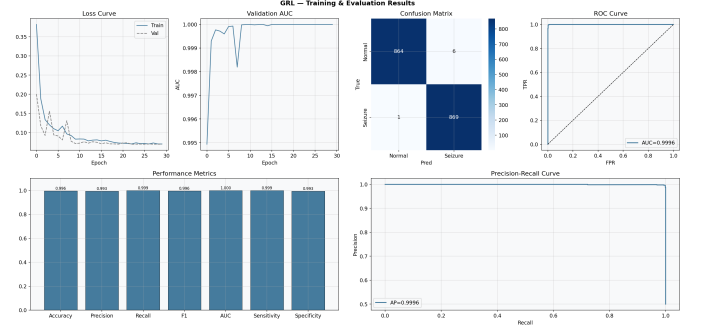


Fig. 11. GRL model training and evaluation results showing loss curves, validation AUC, confusion matrix, ROC curve, performance metrics, and precision-recall curve.

C. DA-SCT Results

Fig. 12 shows the results for the DA-SCT model. Adding domain adaptation to the SCT architecture reduces the false normal predictions compared to the base SCT. The DA-SCT achieved accuracy 0.983, precision 0.992, recall 0.974, F1-score 0.983, and AUC 0.999.

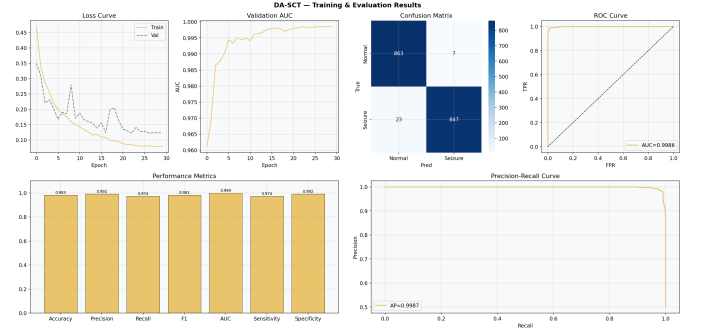


Fig. 12. DA-SCT model training and evaluation results showing loss curves, validation AUC, confusion matrix, ROC curve, performance metrics, and precision-recall curve.

D. DA-GRL Results

Fig. 13 shows the results for the proposed DA-GRL model. The DA-GRL achieves the best results among all model variants. The confusion matrix shows zero false normal predictions, meaning all actual seizure windows are correctly detected. The DA-GRL achieved accuracy 0.997, precision 0.993, recall 1.000, F1-score 0.997, and AUC 1.000. **DA-GRL achieved best overall seizure detection performance.**

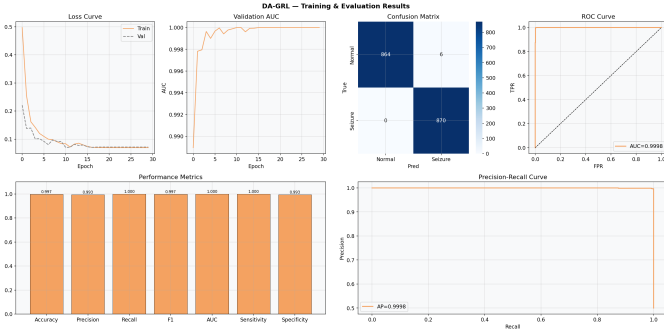


Fig. 13. DA-GRL model training and evaluation results showing loss curves, validation AUC, confusion matrix, ROC curve, performance metrics, and precision-recall curve.

E. Privacy-Preserved Baseline Comparison

For completeness, a privacy-preserved federated learning baseline is included as a reference comparison. This model was trained with differential privacy constraints and simulates a multi-site training scenario. Fig. 14 shows its results. The federated baseline achieved accuracy 0.996, precision 0.992, recall 1.000, F1-score 0.996, and AUC 1.000. While this baseline performs comparably to the centrally trained GRL model, it is not part of the proposed system and is included only as an additional comparison point.

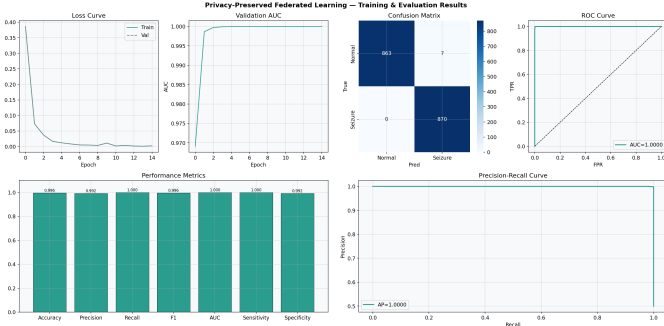


Fig. 14. Privacy-preserved federated learning baseline results included as a comparison reference. This model is not part of the proposed framework.

F. Model Comparison Summary

Table I presents a comprehensive comparison of all four proposed model variants and the federated baseline across all performance metrics.

TABLE I
PERFORMANCE COMPARISON OF ALL MODEL VARIANTS ON THE CHB-MIT TEST SET.

Model	Acc.	Prec.	Rec.	F1	AUC
SCT	0.986	0.990	0.983	0.986	0.999
GRL	0.996	0.993	0.999	0.996	1.000
DA-SCT	0.983	0.992	0.974	0.983	0.999
DA-GRL	0.997	0.993	1.000	0.997	1.000
Fed. Baseline	0.996	0.992	1.000	0.996	1.000

Fig. 15 shows a radar chart comparing SCT, GRL, DA-SCT, and DA-GRL across all five performance metrics. The DA-GRL model covers the largest area on the radar chart, confirming its superior overall performance. The domain-adapted models (DA-SCT and DA-GRL) show improved recall compared to their non-adapted counterparts in most cases.

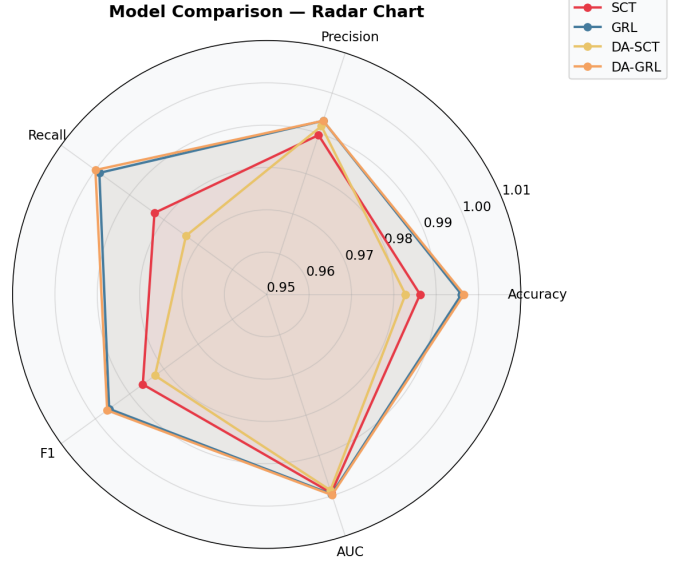


Fig. 15. Radar chart comparing SCT, GRL, DA-SCT, and DA-GRL models across precision, accuracy, AUC, F1-score, and recall. DA-GRL covers the largest area, reflecting best overall performance.

VIII. EXPLAINABLE AI ANALYSIS

Explainability is a key requirement for clinical adoption of AI-based diagnostic tools. This section presents the XAI analysis of the DA-GRL model using SHAP, LIME, temporal analysis, and attention maps.

A. SHAP Channel Importance

SHAP (SHapley Additive exPlanations) values are computed for the DA-GRL model using the GradientExplainer method on 120 test samples [7]. Fig. 16 shows the full channel importance ranking and the SHAP value distribution for the top five channels. The C3-P3 channel is identified as the most important channel with a mean absolute SHAP value of approximately 0.00030. The T8-P8 channel ranks second, followed by F3-C3, F4-C4, and P7-O1. These channels cover the central and temporal-parietal regions of the brain, which are known to be active during focal seizures.

B. Unified XAI Report

Fig. 17 shows the unified XAI report for the DA-GRL model, which brings together channel importance, SHAP distributions, temporal importance profiles, and the seizure versus non-seizure SHAP comparison in a single view. The performance summary confirms accuracy of 0.9966, precision of 0.9932, recall of 1.0000, F1-score of 0.9966, and AUC of 0.9998. The seizure versus non-seizure SHAP comparison

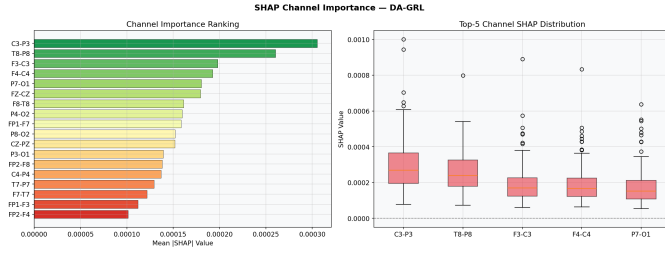


Fig. 16. SHAP channel importance for DA-GRL. Left: full ranking of all 18 channels by mean absolute SHAP value. Right: box plot of SHAP distributions for the top five channels (C3-P3, T8-P8, F3-C3, F4-C4, P7-O1).

shows that seizure windows consistently produce higher SHAP values for the dominant channels, particularly C3-P3 and T8-P8.

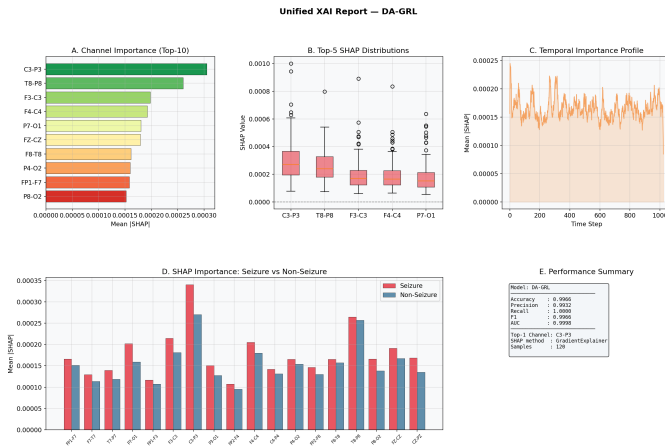


Fig. 17. Unified XAI report for DA-GRL showing: (A) top-10 channel importance ranking, (B) top-5 SHAP distributions, (C) temporal importance profile, (D) SHAP importance comparison for seizure versus non-seizure windows, and (E) model performance summary.

C. SHAP Beeswarm and Temporal Heatmap

Fig. 18 presents two advanced SHAP visualizations. The beeswarm plot on the left shows the distribution of SHAP values for the top ten channels. Each point represents one test sample, and the color indicates the feature value. The plot shows that C3-P3 and F3-C3 channels have the widest spread of high SHAP values, confirming their strong influence on model decisions. The temporal SHAP heatmap on the right reveals which time windows within a segment are most important for each of the top five channels. Concentrated areas of high positive SHAP values (shown in red) correspond to seizure-sensitive time windows, particularly around time steps 30–50 and 80–90 for the C3-P3 channel.

D. SHAP vs LIME Comparative Analysis

Fig. 19 presents a full comparative report between SHAP and LIME explanations for the DA-GRL model [7]. SHAP operates as a global and local explainer using white-box gradient access, while LIME operates as a local black-box

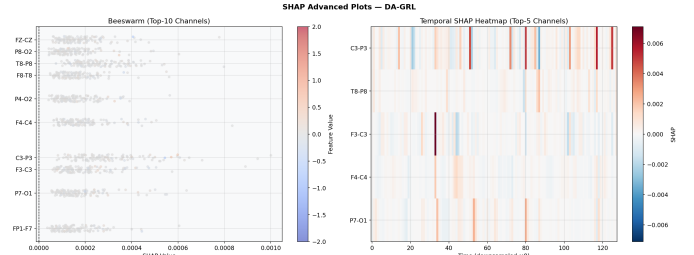


Fig. 18. Advanced SHAP visualizations for DA-GRL. Left: beeswarm plot showing SHAP value distributions for top-10 channels with feature value coloring. Right: temporal SHAP heatmap showing seizure-sensitive time windows for the top-5 channels.

explainer through perturbation. The top-ranked channel by SHAP is C3-P3, while LIME identifies FP1-F7 as the top channel. The Jaccard similarity for the top five channels is 0.1111, indicating moderate but limited overlap. The channel ranking comparison plot shows that SHAP provides a more detailed and differentiated ranking while LIME ranks channels with lower variation. The overall agreement level is assessed as moderate, and the combination of both methods provides a more complete picture of model behavior than either method alone.

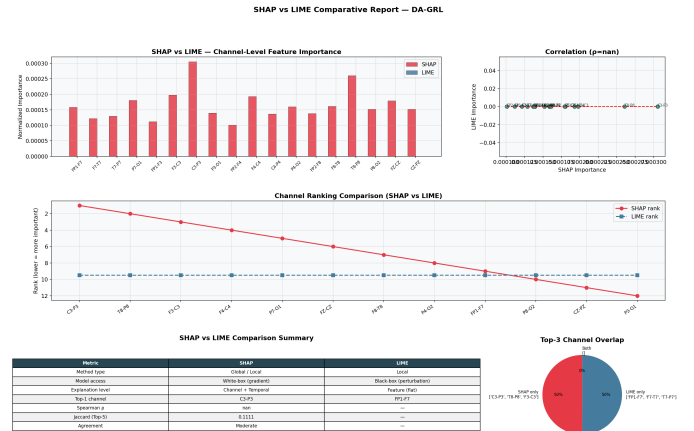


Fig. 19. SHAP vs LIME comparative report for DA-GRL showing channel-level feature importance comparison, correlation scatter plot, channel ranking comparison, comparison summary table, and top-3 channel overlap analysis.

E. LIME Local Explanation

Fig. 20 shows the LIME explanation for a single test sample from the DA-GRL model. The sample is a normal EEG window, and the model correctly predicts it as normal with 95.1% probability. The feature contribution bar chart shows both positive (red) and negative (blue) contributions from individual channel-time features. The most positive contributor is the feature ch_10 at time step 693, while the most negative contributor is ch_1 at time step 318. This sample-level explanation demonstrates that LIME can confirm the local consistency of model decisions.

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